

GreenLawAI: Forestry and Climate Awareness System

Progress Update: January 2026.
Data Pipeline & Logic Layer complete.

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Our 3-Pillar Engineering Approach



DATA

FOUNDATION

Turning messy legal OCR into structured, searchable data

SEARCH

THE BRAIN

Building a Hybrid Search Engine

PRECISION

ACCURACY

Implementing mathematical guardrails to ensure accurate facts

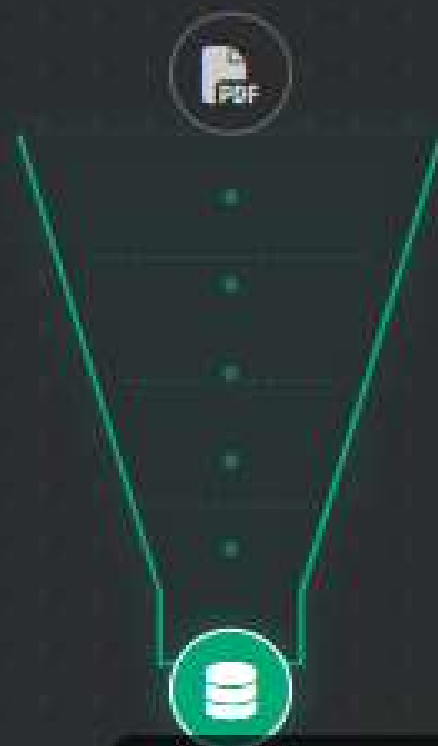


Exceeds target height by 2% *Suggest reducing content or adjusting layout*

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Turning Messy Law into Deep Data

PHASE 1



01

Extraction

Pulling raw text from old scanned PDF's using OCR technology.

02

Cleaning

Removing OCR noise, headers, footers, and formatting artifacts.

03

Segmenting

Detecting Chapters, Sections, and Clauses automatically based on legal structure.

04

Entity Extraction

Finding key legal terms (e.g., Specific Trees, Penalties, Jurisdictions).



Exceeds target height by 5% *Suggest reducing content or adjusting layout*

Categorizing text chunks as "Declarations," "Prohibitions," or "Procedural Rules."

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Semantic Segmenting (The Context)

01 STRUCTURE AWARENESS

AI now knows the difference between Chapters, Sections, and Schedules. It understands the document skeleton.

02 "HOME ADDRESS"

Every paragraph has a unique locator ID. We trace every clause back to its parent Act and Chapter.

03 ZERO HALLUCINATION

Result: AI never gets lost. It cites exactly where the information came from within the legal hierarchy.

HIERARCHY VISUALIZATION



HIERARCHICAL DRIZZLING // DEPTH: 3 LEVELS

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Why one search engine wasn't enough

1 Similarity Brain FAISS

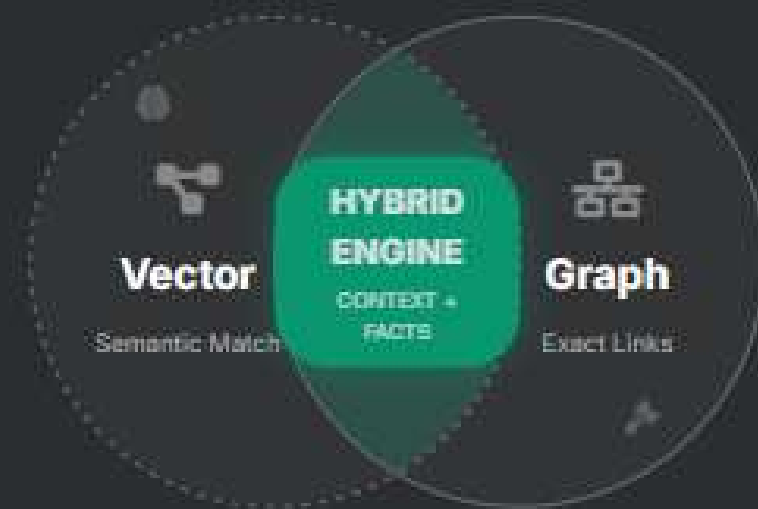
Vector search finds related concepts (e.g., "cutting wood" ≈ "timber harvesting") even if keywords don't match.

2 Connection Brain NEO4J

Graph search understands explicit relationships. It knows that "Section 26" specifically links to "Penalty Clause B."

THE RESULT

A system that "reasons" with context rather than just matching keywords.



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EKO(Expert Knowledge Overlay)

01 We hard-coded verified "Fact Chunks" for critical laws to bypass probabilistic errors.

02 System **prioritizes** these immutable facts above all other AI-generated search results.

03 Result: **100% accurate fines.**
(e.g., Deodar Tree Penalty = Rs. 206,000)

VERIFIED

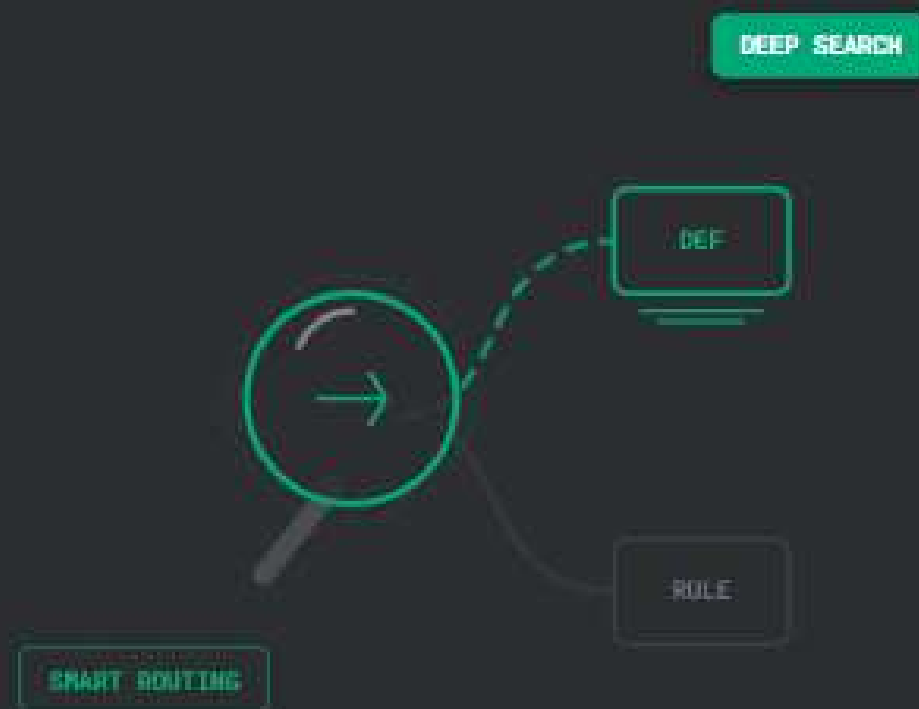


IMMUTABLE LAYER

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Thinking Before Searching

- 01 The system guesses if you want a "Definition" or a "Rule" before executing search.
- 02 For definitions, it automatically digs 20x deeper into the vector database layers.
- 03 It proactively blocks "circular" logic (e.g., preventing the system from defining a word by itself).



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Local Speed + Cloud Intelligence

LOCAL Student Laptop

All sensitive data is **stored and indexed** locally. The search engine (FAISS) runs on-device for speed.

CLOUD Google Colab

We send only the retrieved "context" to the cloud. The **Llama 3 Model** generates the final answer remotely.

RESULT Performance

Professional-grade AI reasoning running smoothly on a **standard student laptop**.



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Precision Milestones



Identifies Department Hierarchy

Correctly maps reporting lines from Forest Guard to Conservator.

Verified



Cites 2022 Forest Amendments

Automatically retrieves the latest updated statutes, ignoring repealed acts.

Verified



Suppresses Circular Logic

Prevents recursive definitions for ambiguous terms like 'Timber' and 'Trees'.

Verified

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Final Delivery Goals

GOAL 01

Logging Monitor

Building a real-time dashboard to visualize deforestation trends and report generation statistics.

GOAL 02

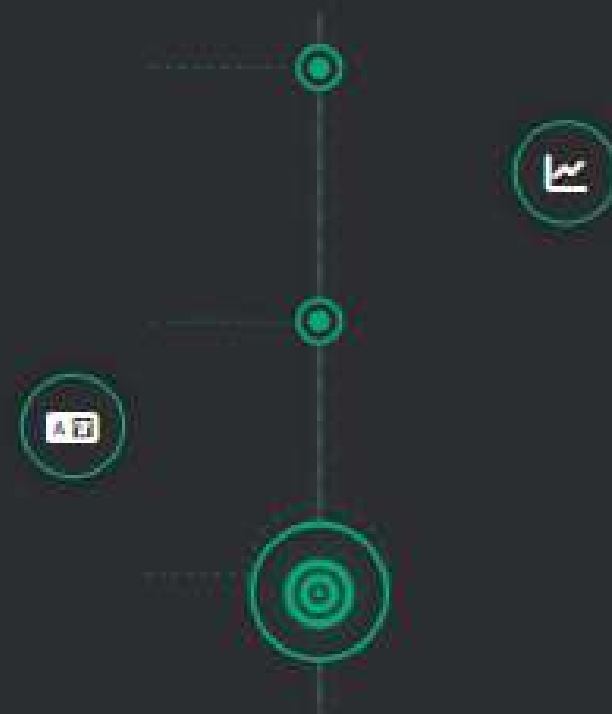
Urdu Language Support

Integrating multilingual translation models to make forest laws accessible to local guards.

GOAL 03

System Deployment

Final containerization (Docker) and documentation release for university submission.



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